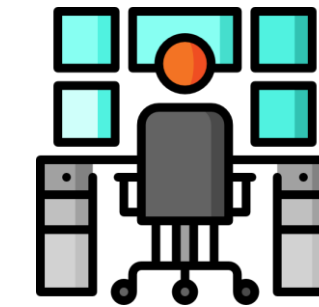




CODENOVATE



Vigilance: Real-Time Monitoring for Proactive Events

{OPEN INNOVATION}

TEAM NAME: Ctrl+Include

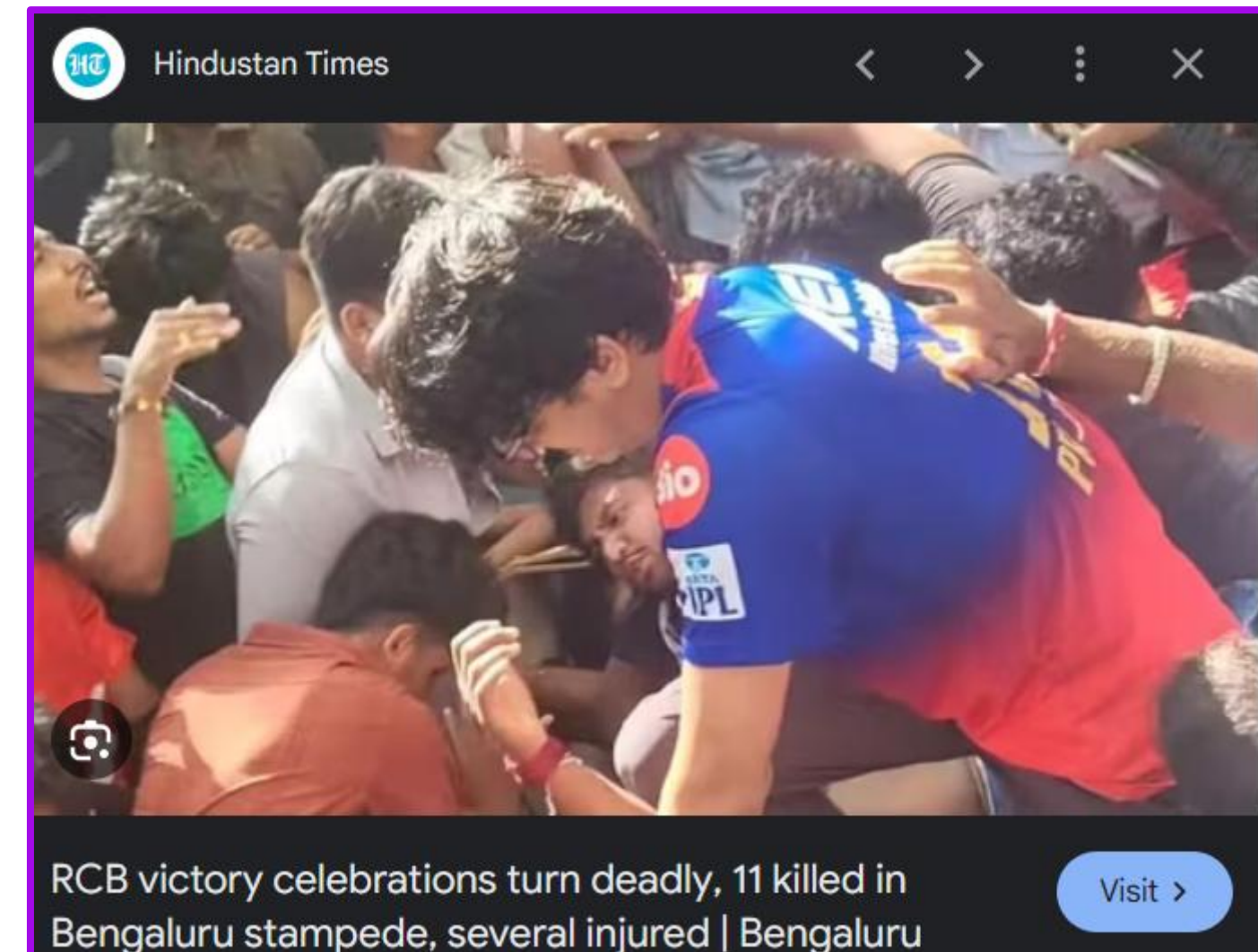
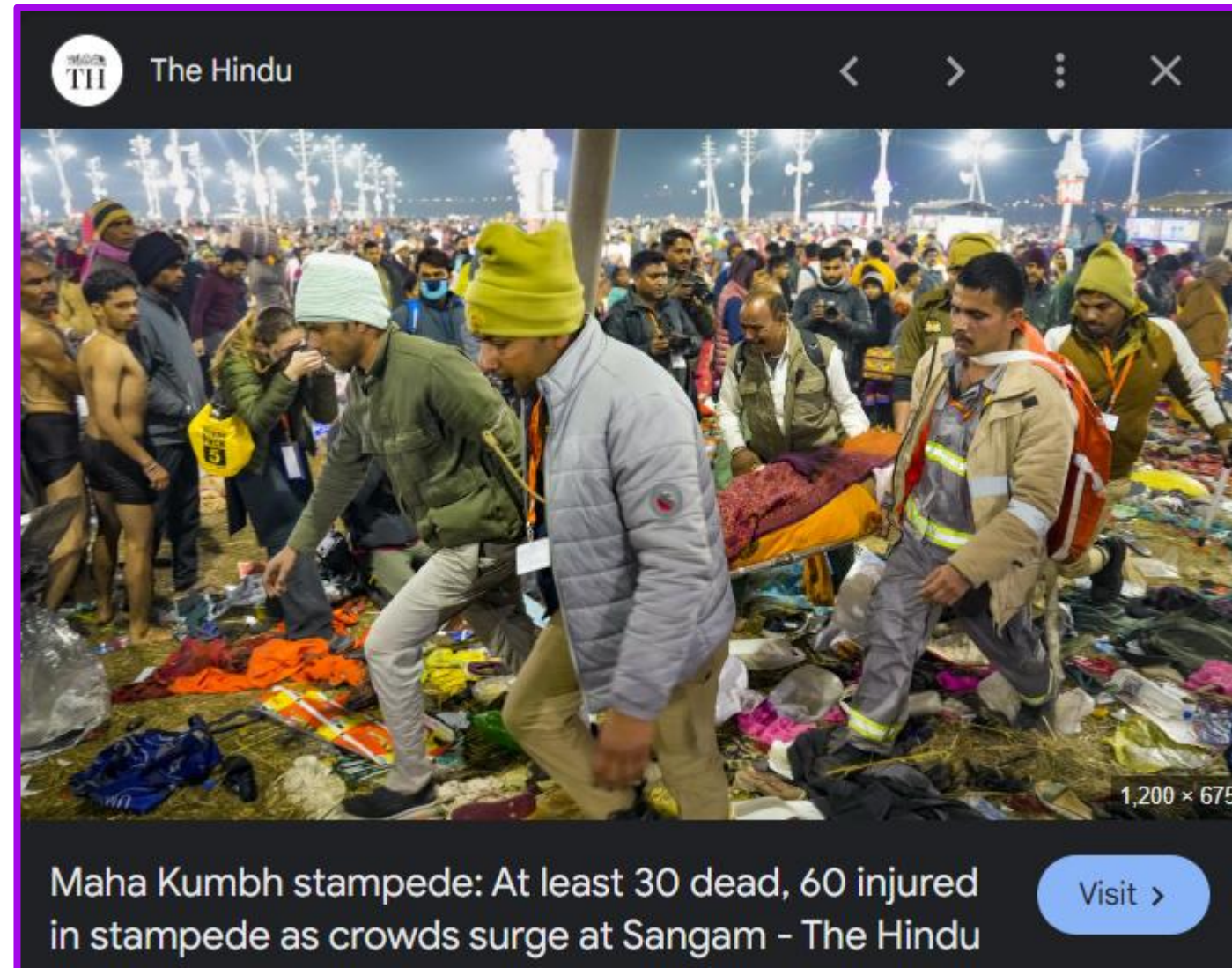
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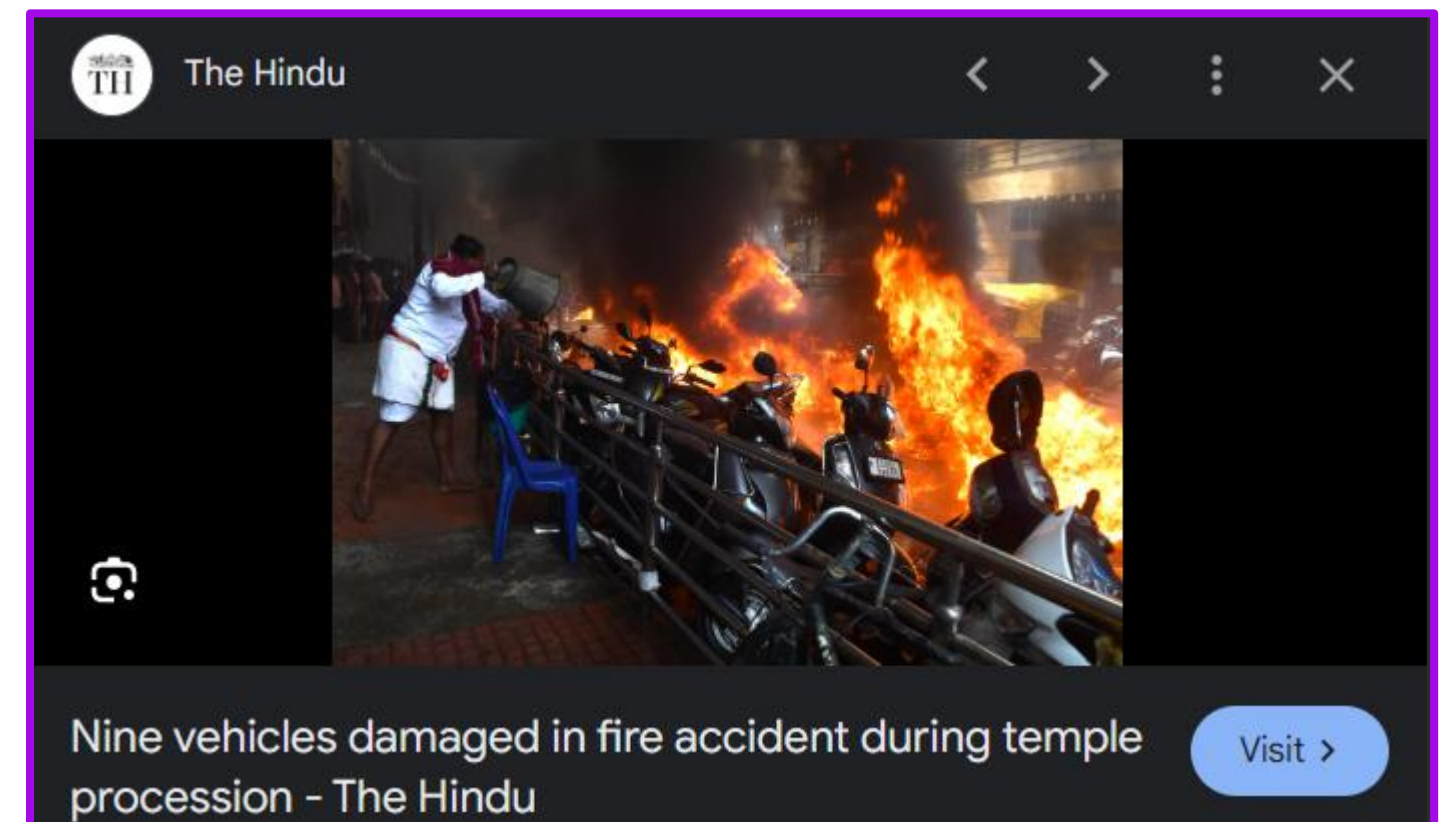
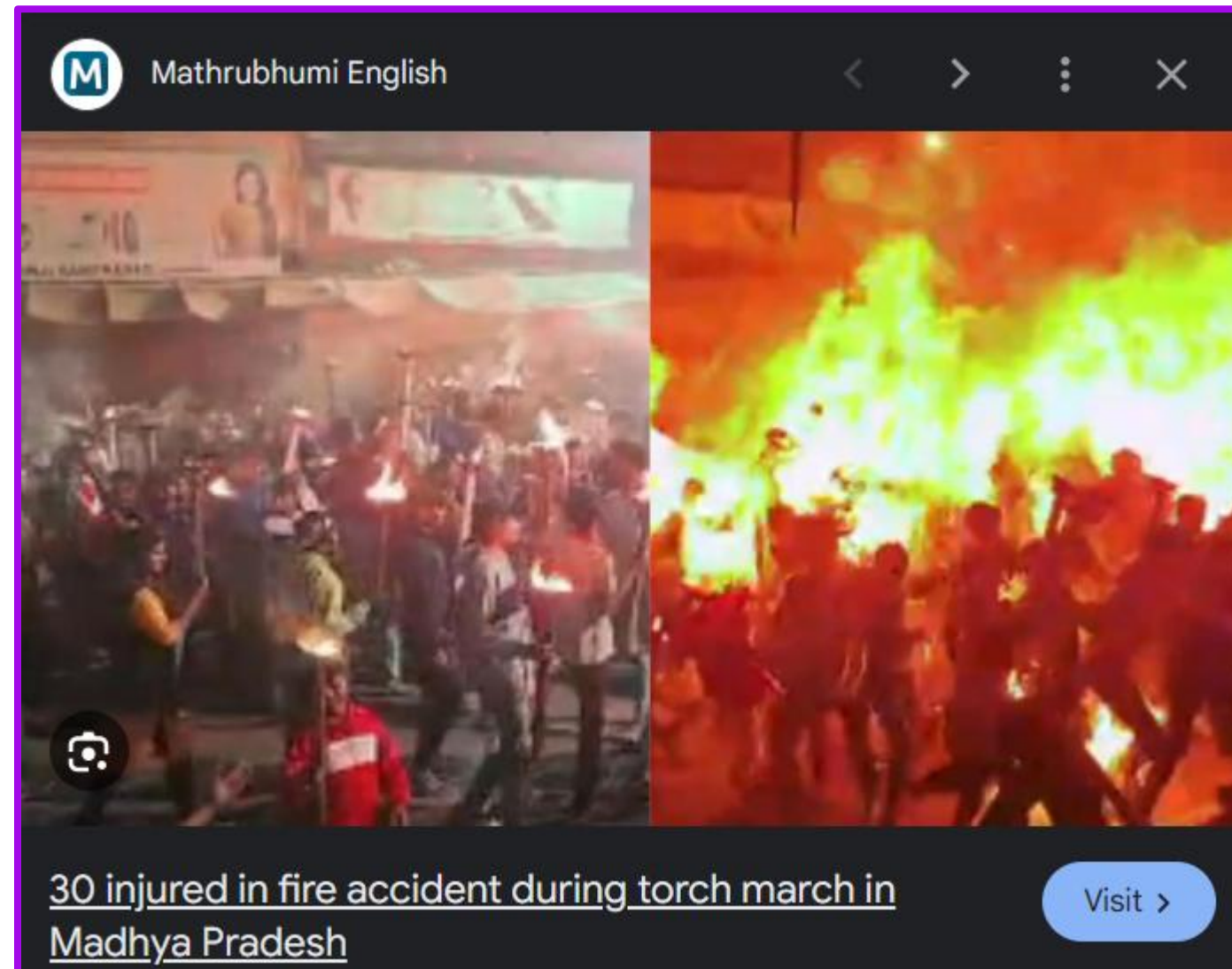
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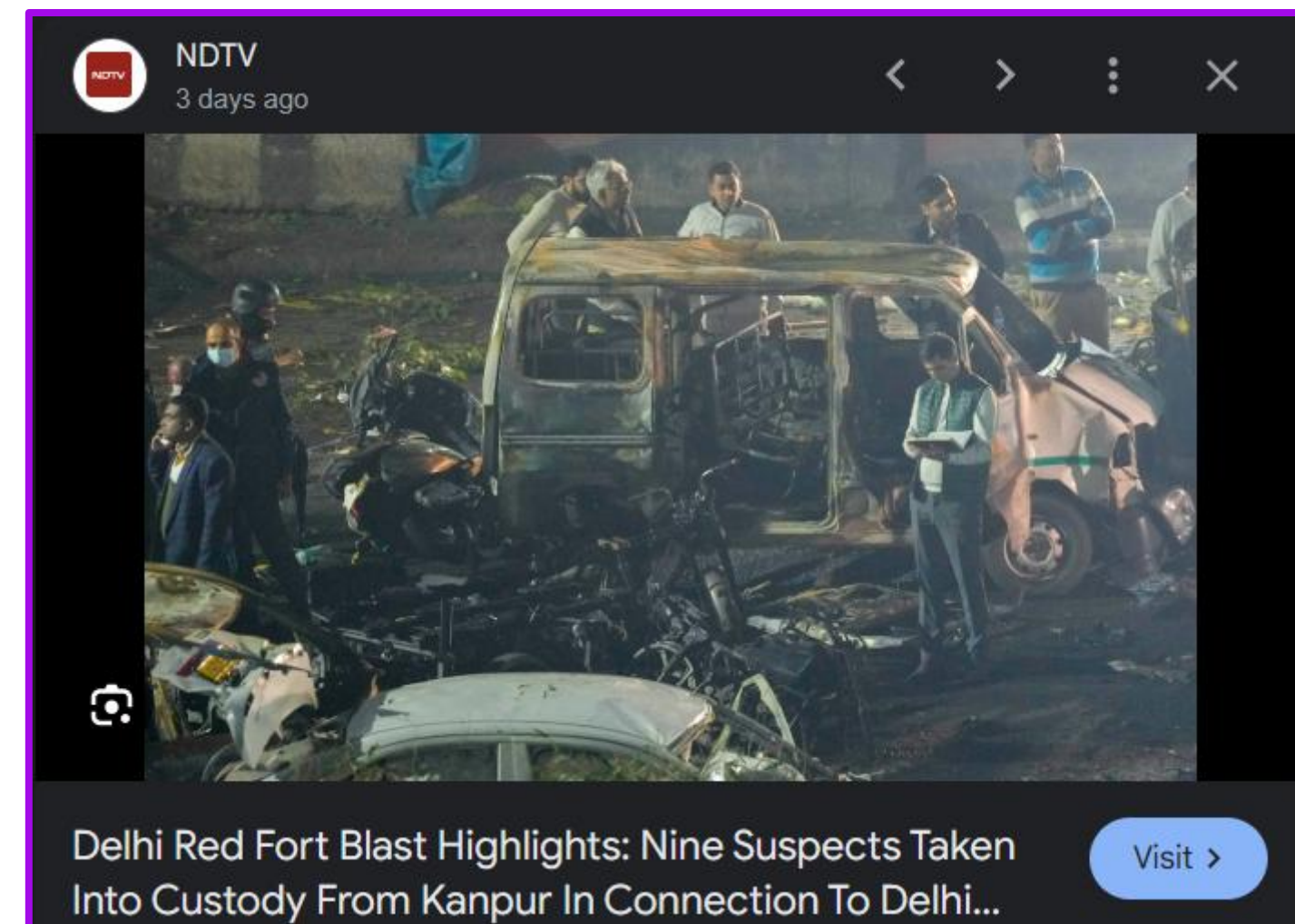
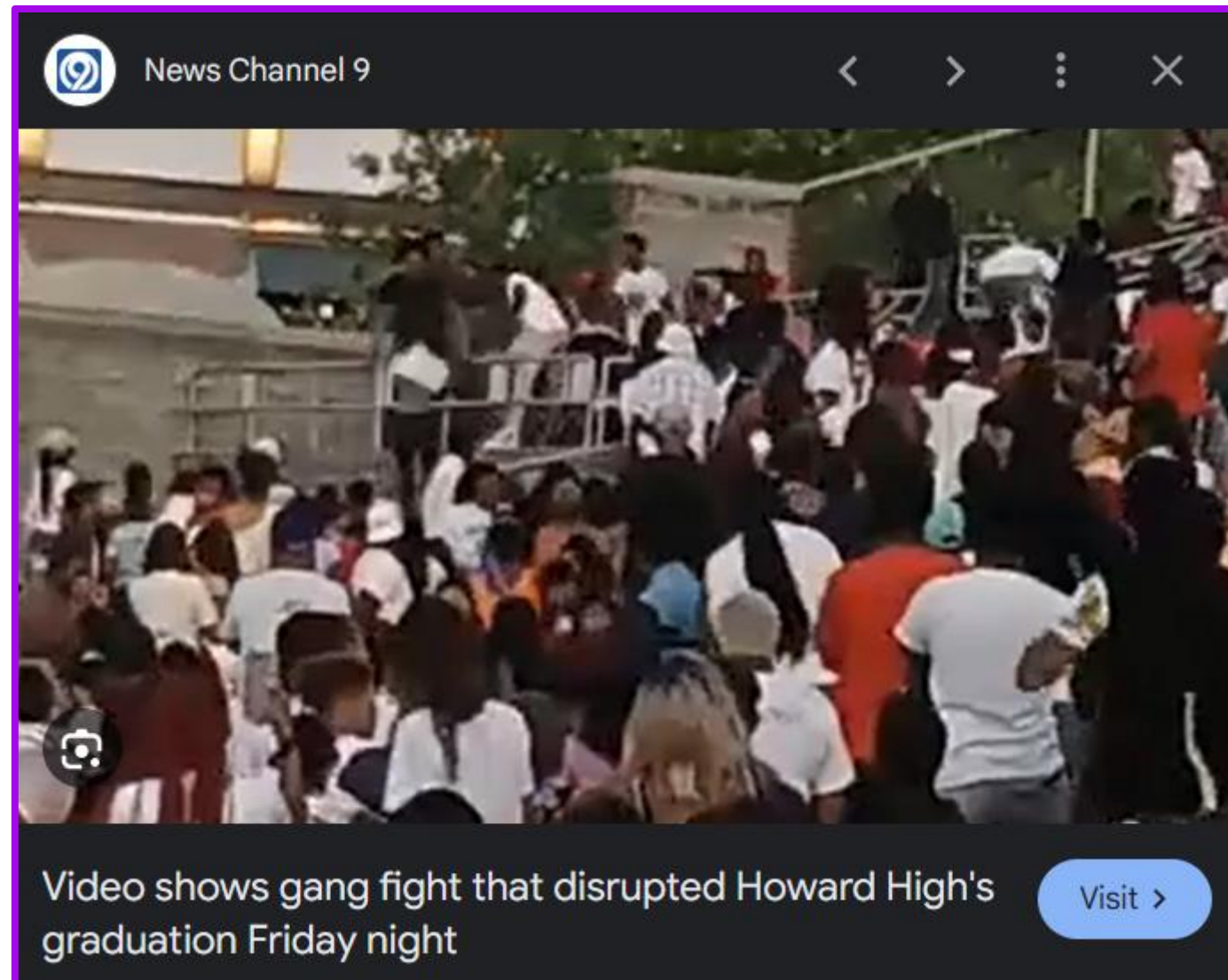
CROWD SURGE/ STAMPEDE



FIRE BREAKOUTS



VIOLENCE + FIGHTS



PROBLEMS

- MISSING CHILDREN
 - LOSING PERSONAL BELONGINGS
 - SUSPICIOUS ITEMS (BANNED ITEMS: GUN)
 - CONFUSING DIRECTIONS
 - WEATHER ISSUES
 - MANAGEMENT ISSUES (EVENT SCHEDULE)
 - COMMUNICATION ISSUES (ANNOUNCEMENTS NOT HEARD CLEARLY)
 - COORDINATION ISSUES
 - CROWD SURGE
 - FIGHTS
 - FIRE BREAKOUTS
- ✓ concerts
 - ✓ sports events
 - ✓ college fests
 - ✓ political rallies
 - ✓ exhibitions
 - ✓ religious gatherings

Problem Statement

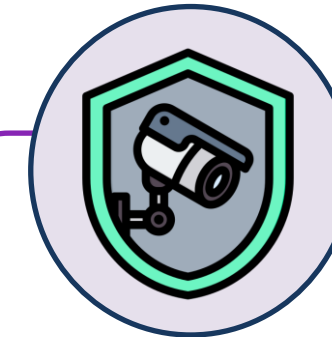
Large-scale public events (like concerts, rallies, or sports gatherings) face significant challenges in **real-time safety, coordination, and operational efficiency**. Current monitoring systems **rely on human oversight**, making them **reactive, error-prone, and unable to keep pace** with massive crowds and dynamically evolving scenarios. This critical gap results in **substantial operational risk** due to **fragmented, uncoordinated responses** and **critical delays** in managing threats like stampedes, medical emergencies, fire outbreaks, or security issues, demonstrating an urgent need for intelligent systems that enable **proactive safety management**.

Solution

Vigilance— Proactive Incident Management

Vigilance is an **integrated, product-grade platform** designed for **proactive and intelligent incident management**. It addresses the shortcomings of traditional systems by leveraging **advanced AI, computer vision, and geo-intelligence** to continuously monitor the event space. The platform's core purpose is to **enhance safety, optimize resource allocation, and ensure reliable, synchronized response** to all critical situations at major events.

Vigilance Modules



Vision

(Live Camera Feed Analysis)

- **AI-powered computer vision** monitors video streams in real time, detecting critical events like
 1. Fire & smoke
 2. crowd surges
 3. Violence
 4. Thermal anomalies
- Incidents are **geo-located** and trigger **instant alerts** for coordinated security response.



Locator

(AI-Powered Matching & Recovery)

- Uses **AI-powered matching** for both items and people.
- For **lost items**, it uses **visual similarity** and data matching.
- For **missing persons**, it uses **facial recognition** to find matches and directs **nearby volunteers** for safe reunification.



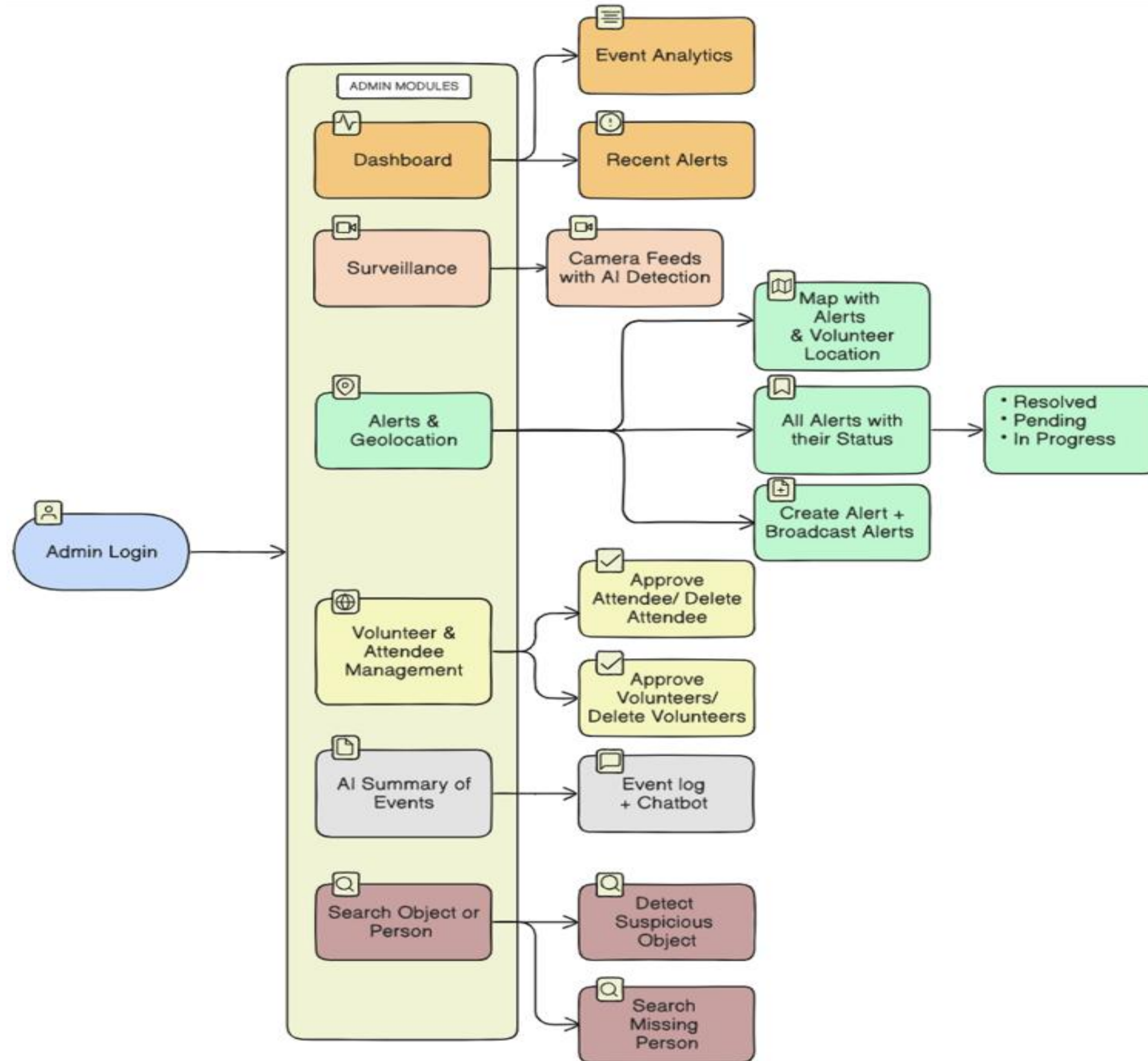
Navigator

(User Role-Based & Geo-Intelligence)

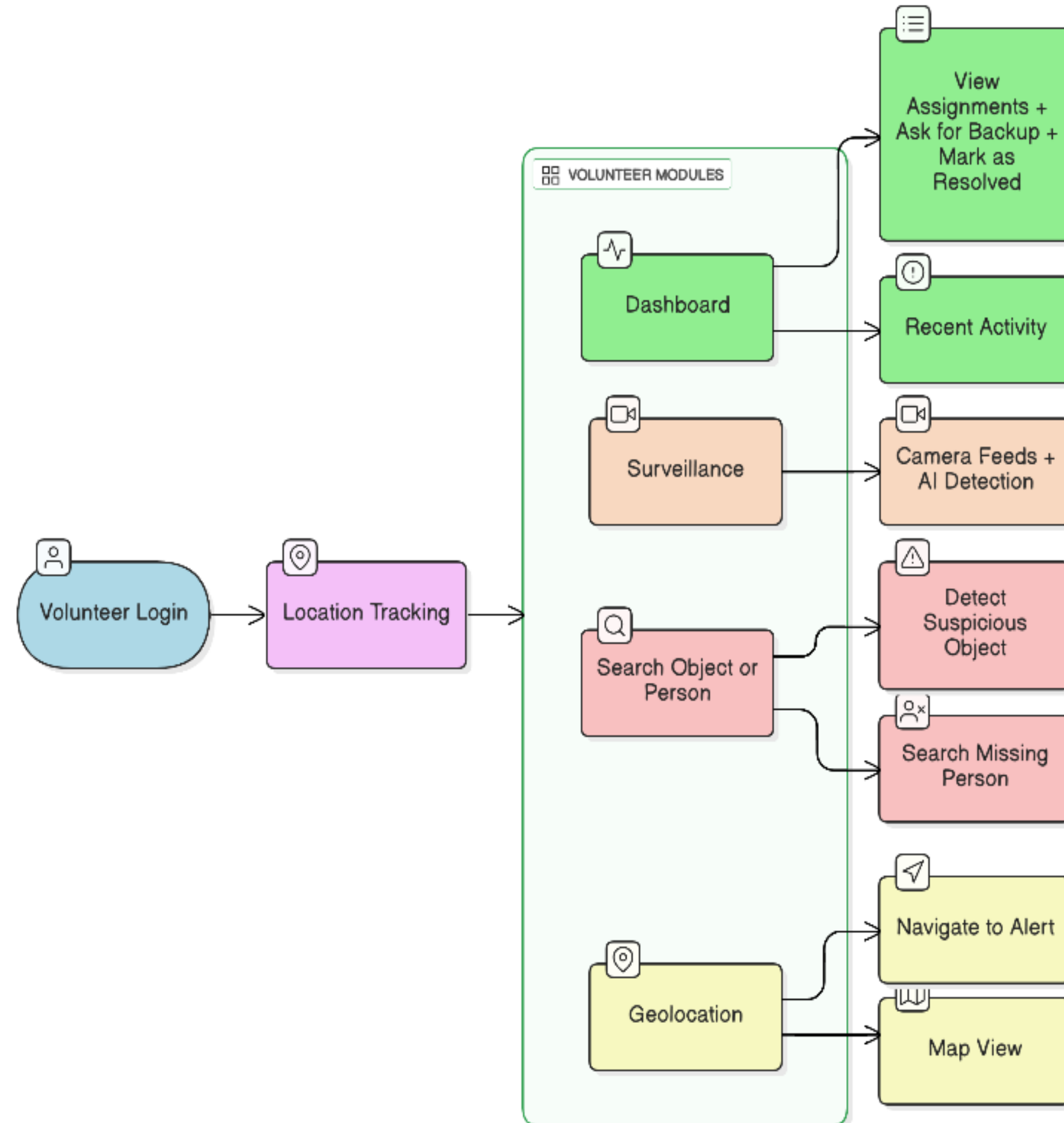
- The system provides **role-based access** for
 1. **Admins** : (complete oversight, dispatch etc.)
 2. **Volunteers** : validated alerts/navigation
 3. **Attendees** : (safety updates/nearby exits).
- All modules are **GPS-synced** for real-time tracking, optimized routing, and dynamic emergency evacuation.

HITL (Human-in-the-Loop): Ensures critical AI-detected incidents are verified by humans for accuracy and reliability before action.

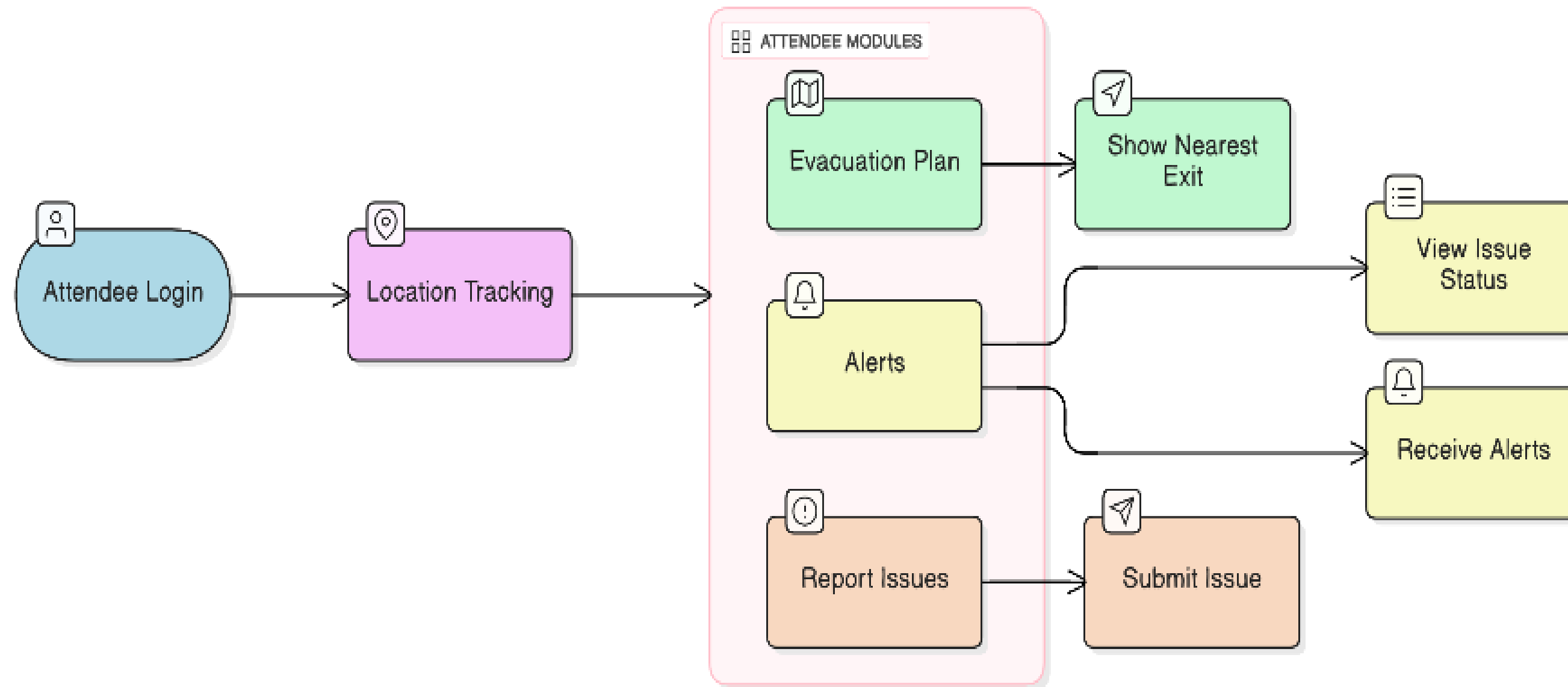
ADMIN



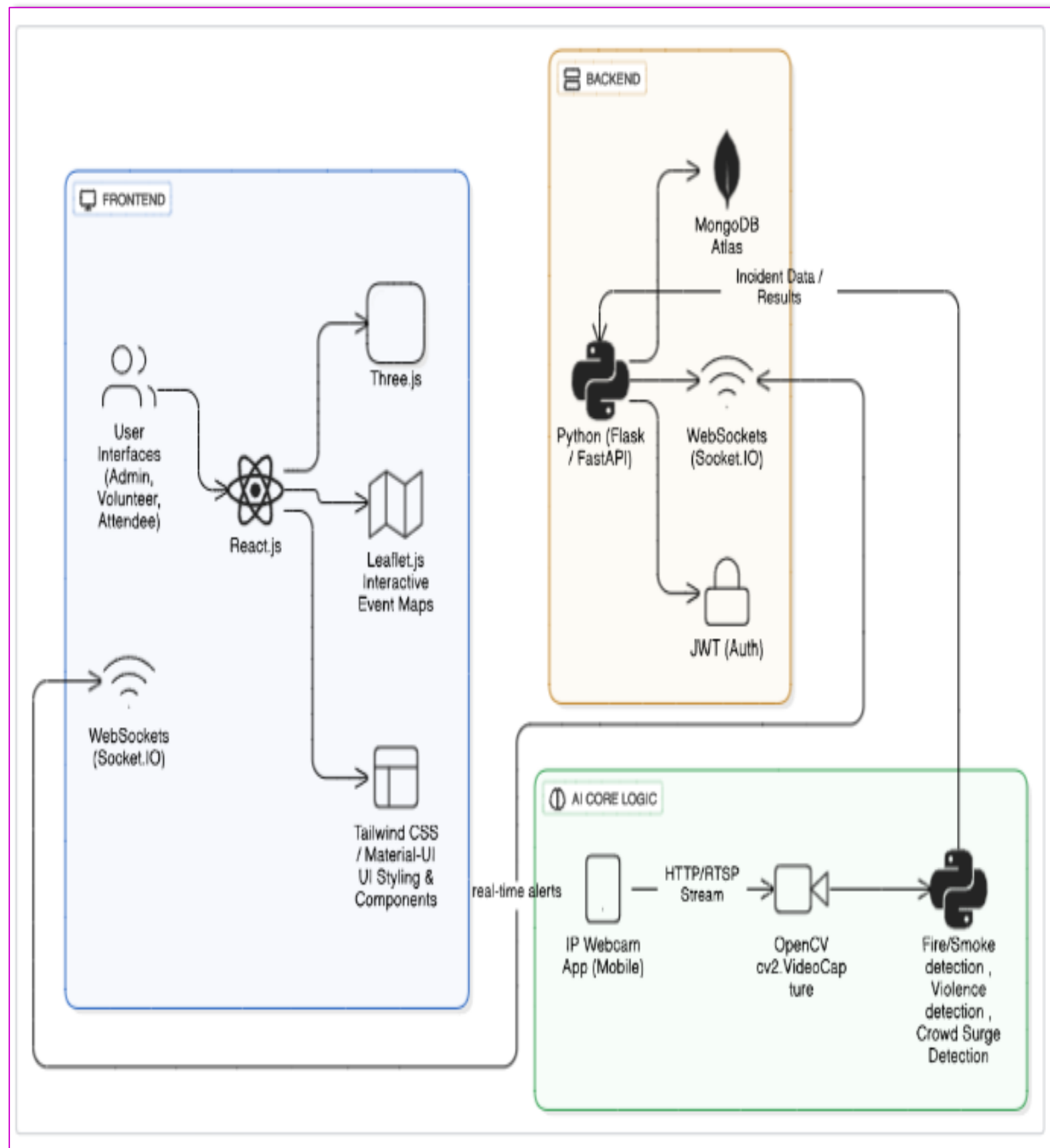
VOLUNTEER



ATTENDEE



WORKFLOW



TECH STACK

Frontend

React.js / Three.js: Interactive Dashboards & 3D Map Rendering.
Leaflet: Geo-location & Real-time Route Mapping.
WebSockets (Socket.IO): Instant Alert & Data Updates.



Backend

Python (FastAPI): Core Business Logic Server.
MongoDB Atlas: Scalable NoSQL Database for all event data.
JWT: Secure, Role-Based Authentication.



AI - Vision

TensorFlow & PyTorch & Keras: Core ML Framework hosting **specialized models for:**

- **Object Detection:** Fire/Smoke & Violence Detection.
- **Crowd Analysis:** Density Estimation / Surge Detection.
- **Similarity Search:** Face & Lost Item Matching.
- **OpenCV (cv2):** Video Stream Ingestion and Frame Processing.



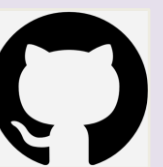
Streaming (MVP)

IP Webcam App → OpenCV: Low-latency camera feed ingestion.



DevOps & Deployment

Docker: Simple Containerization.
Render & Vercel: Cloud Hosting.



BUSINESS SCOPE

Data Monetization & Insights –

Offer anonymized crowd behavior analytics to event planners, advertisers, and smart city departments for better planning and marketing.

API & White-Label Partnerships –

License your AI models to third-party event management platforms, CCTV vendors, and smart city applications

Private Security Firms – Security Contractors managing safety at VIP or corporate events.

Partnership with Security & Event Agencies –

Collaborate with event organizers, police, and emergency response teams to provide integrated safety solutions.

Cross-Domain Expansion –

Extendable beyond events to airports, campuses, transportation hubs, and smart city surveillance.

Smart Volunteer Ecosystem – Build a verified volunteer and responder network accessible through the platform, enabling paid partnerships with event organizers.

Innovation vs Existing Approaches

Key Differentiator	Manual Monitoring (CCTV)	Basic Surveillance Systems	Traditional Management Tools	Our Solution (AI Event Monitor)
1. Real-Time Incident Detection	✗	✓	✗	✓
2. Geo-Intelligent Volunteer Routing	✗	✗	✗	✓
3. Facial Recognition & Role-Based Access	✗	✓	✗	✓
4. Multi-Device & Cross-Platform Support	✗	✗	✓	✓
5. Predictive Analytics & Risk Forecasting	✗	✗	✗	✓
6. Automated Alerts & Crowd Insights	✗	✓	✗	✓
7. Integrated Dashboard for Real-Time Control	✗	✓	✓	✓
8. Scalable & Cloud-Based Architecture	✗	✗	✓	✓

SOCIAL IMPACT

Scalable Public Safety Infrastructure – The system can evolve into a smart-city safety backbone, disaster management and security



Social Good & Inclusivity – Builds public trust, ensures family safety, and showcases technology's positive impact on communities.

Cost-Efficient – Reduces manpower needs and operational costs through automation and analytics.



Geo-intelligent routing guides volunteers for faster, data-driven action, enabling smart navigation and efficient response.



Facial recognition and role-based access enhance system security, reliability, and user authentication.



Economic & Brand Value Boost – Safer, well-managed events attract more attendees, sponsor trust, and higher profitability.



Challenges

Real-Time Processing Load – Handling multiple live video streams simultaneously with minimal latency and high accuracy.

Accurate Anomaly Detection – Differentiating between normal crowd movement and genuine emergencies like panic or violence.

Low-Quality Video Inputs – Dealing with blurry, low-light, or lagging feeds that make AI analysis less reliable.

Network Reliability – Maintaining system stability even under weak connectivity or overloaded communication networks.

User Adoption & Training – Encouraging volunteers, security teams, and organizers to effectively use the platform in real-world conditions.

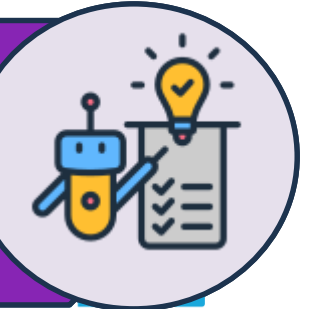
Future Scope

Public Accessibility Mode – Provide a simplified version for differently-abled attendees, including voice guidance and visual safety cues.



Device-Independent Operation – Delivers consistent performance and access across multiple platforms without hardware limitations.

AI Ethics & Transparency Framework – Finetuning AI models to ensure fairness, accountability, and trust in automated decisions.



Offline Mode Functionality – Enable local processing for continuous safety monitoring even without internet access.

Predictive Safety Intelligence – Use AI trends and historical data to forecast crowd risks and prevent incidents before they occur.



Incident Validation through Cross-Camera View – Use multi-angle camera verification to confirm real incidents and reduce false alarms.

Conclusion

Our AI-powered Event Monitoring System enhances safety, accessibility, and efficiency through real-time detection, geo-intelligent routing, and facial recognition. It transforms traditional surveillance into proactive, data-driven management — ensuring faster response, optimized resources, and inclusive event operations. This innovation bridges technology with social good, setting a new standard for smart, scalable, and secure public systems.

THANK YOU



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